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Optimizing Performance of Hydrogen Underground Storage in Aquifers by Evaluating the Impact of Key Operational Design Parameters

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Abstract

Seasonal storage of hydrogen (H_2) in saline aquifers is a viable technology for large-scale hydrogen energy storage. Numerical simulation of hydrogen underground storage (HUS) in porous media is a promising technique for evaluating and designing such process. The main objective of this study is to optimize the performance of the HUS process by investigating the impact of key design parameters, such as cushion gas (CH_4, N_2 and CO_2) requirements and utilization; H_2 injection and production cycles; and well completions and placement, on storage capacity, reservoir pressure, produced H_2 purity and hydrogen recovery efficiency.

Multiple H_2 gas injectors/producers were placed at the crest of the structure to utilize most of the available pore volume. Various operational schemes of the HUS process were selected, which started with injection of cushion gas followed by multiple cycles of hydrogen injection and production. Several sensitivity cases were developed considering injection of three different types of cushion gas (CH_4, N_2 and CO_2) in different volumes and different lengths of H_2 injection and production periods with multiple well configuration options. The impact on key HUS design parameters, such as storage capacity, reservoir pressure, produced H_2 purity and hydrogen recovery efficiency, were evaluated.

Results showed that injection of cushion gas, prior to H_2 injection/production cycles, is needed to build reservoir pressure, which ensures H_2 deliverability. Regardless of the cushion gas type, H_2 recovery efficiency increases with increasing the cushion gas-hydrogen ratio (CGR); however, beyond CGR of 1.5, there is no benefit in this case. Lighter cushion gas (N_2 and CH_4) injection provided better recovery efficiency than CO_2 due to less gas segregation and viscous fingering. Hydrogen wells placed updip and completed in the top part of the reservoir resulted in better performance in terms of H_2 recovery and purity in all types of cushion gas scenarios due to reduced cushion gas back production and less water encroachment. An operational scheme, including H_2 injection and withdrawal rates and length of injection/production period, should be optimized to achieve higher H_2 recovery efficiency.

This study provides not only the comprehensive evaluation of cushion gas requirements but also highlights the importance of operational design parameters, which are essential in designing and optimizing performance of HUS projects.

Introduction

The world is currently in a transitional phase from traditional fossil fuel-based energy sources, such as (oil, gas, coal, etc.) to lower carbon and renewable sources of energy production (e.g., geothermal, wind, solar, hydrogen, etc.) with the aim of reducing carbon emissions, which ultimately will lower climate change and its after effects. Hydrogen (H_2) is considered to be one of the key components in this energy transition toward a low-carbon economy (McPherson et al., 2018) [1]. It can play an essential role in tomorrow's energy mix as it can decarbonize all economic sectors and industries, from fueling vehicles to generating electricity, petrochemical industry (ammonia production, methanol production as well as petroleum refining, etc.). Complementing other decarbonization technologies, hydrogen has a central role in helping the world to reach net-zero emissions by 2050 and limit global warming to 1.5 degrees Celsius (Hydrogen Council, 2021) [2].

Since renewable energy sources (e.g., wind power and solar energy) undergo seasonal fluctuations due to atmospheric changes, there are always periods of excess and shortfalls of produced energy. Likewise, the usage of natural gas is cyclic, due mainly to increased heating requirements during winter months. Therefore, this excess renewable energy and natural gas can be converted into hydrogen and stored so that it can be used during high-demand seasons or whenever needed (Noussan et al., 2021) [3].

Storage of hydrogen in surface tanks/containers and salt caverns has been practiced for many years; however, due to their limited storage and discharge capacities, they are suitable for short- to medium-term energy demand fluctuations (Wallace et al., 2021) [4]. To provide energy at larger scale, i.e., in GWh/TWh for weeks or months and hence to have a decarbonization impact at the industry or regional levels, HUS in porous media, such as in saline aquifers, is needed. Hematpur et al. (2023) [5] have reported a detailed review of ongoing large-scale prospective and successful HUS projects worldwide, suggesting future work and gaps in our current understanding.

Hydrogen is the lightest known element on earth; therefore, its storage in porous media has several challenges. Despite the significant potential of HUS in aquifers, the maturity level of the process itself is considered low due to several associated uncertainties and challenges, such as H_2-cushion gas interaction, H_2 plume containment, and in-situ geochemical and microbial reactions, etc. (Heinemann et al., 2021) [6] [7]. In a typical HUS project, H_2 is injected and withdrawn from the aquifer in a cyclic manner depending upon H_2 supply and demand, as shown in Figure 1 [8]. Cushion gas, such as CO_2, CH_4, or N_2, is usually injected prior to H_2 cycles to maintain the reservoir pressure and production flow rate. However, back-production of injected cushion gas may lower the purity of produced hydrogen and its recovery efficiency. Production of brine from saline aquifers also contributes to the purity reduction of produced hydrogen.

Hydrogen underground storage in saline aquifers is a developing technology, and has been actively studied during the past few years. There are several recently published studies in the literature (e.g., Okoroafor et al. [9]; Feldmann et al. [10]; Zamehrian et al. [11]; and Kanaani et al. [12]) investigating the HUS performance using reservoir simulations. However, there is some disparity among the reported simulation results primarily due to different input for key design parameters such as reservoir structure, reservoir pressure and temperature conditions, well placement and completion, cushion gas type and operational schemes. Considering the complexity in the assessment of HUS performance, the main purpose of this work is to perform a comprehensive evaluation at the full-field level, incorporating multiple scenarios of key HUS operational design parameters. It, therefore, attempts to answer three principal questions regarding the storage of hydrogen in finite saline aquifers: 1) What is the performance of each type of recommended cushion gas and what is the optimum volume of each cushion gas required in the form of cushion gas to hydrogen ratio (CGR). 2) How the well placement at different reservoir locations and completion interval can affect hydrogen recovery efficiency and produced hydrogen purity. 3) How the operational scheme, which includes the number of cycles, H_2 injection/production period lengths and H_2 injection and withdrawal rates can be important in designing a successful HUS project.

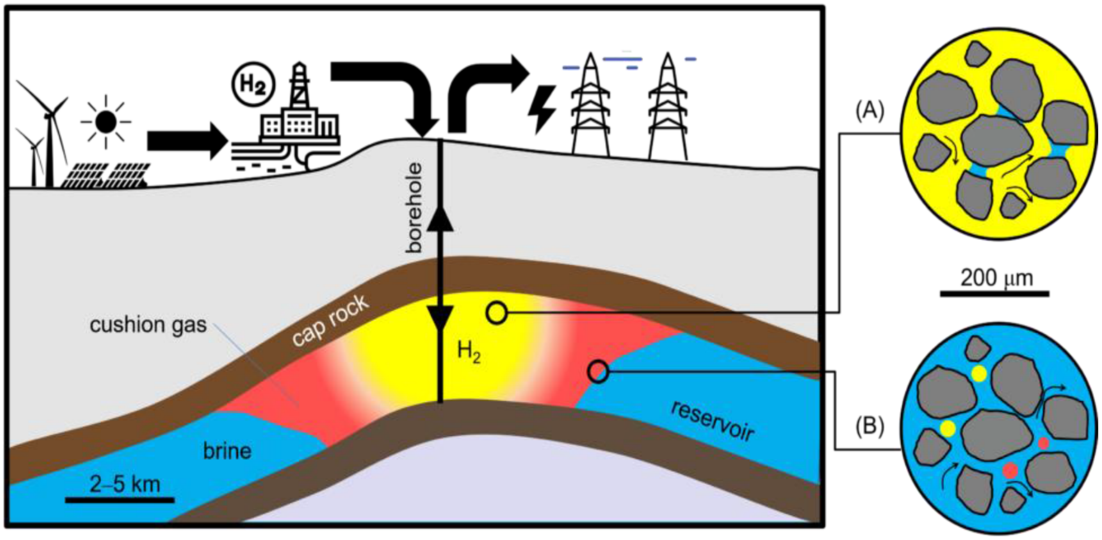

Figure 1—The conceptual illustration of H_2 underground storage in a saline aquifer. (A) shows the ideal H_2 plume displacement of mobile formation water. (B) shows H_2 bubbles trapped and immobilized by capillary forces *© Anozie Ebigbo, Universität der Bundeswehr Hamburg*

Methodology

In this work, a 3D numerical model of an anticline structure, representing a finite aquifer, was built to perform simulation studies related to hydrogen underground storage using synthetic data. The aquifer structure is 25 km in length and 18 km in width, and has a thickness of 100 m. The grid cell dimensions of the reservoir model were 502 in X-direction, 379 in Y-direction and 60 in Z-direction. The total number of grid cells in the model was approximately 11.4 million.

Model grid properties, like porosity and permeability, were generated by using random heterogeneity seeds. Normal distribution was used for porosity values of the population while permeability distribution (Figure 2) was assumed to be correlated with porosity using log-normal distribution.

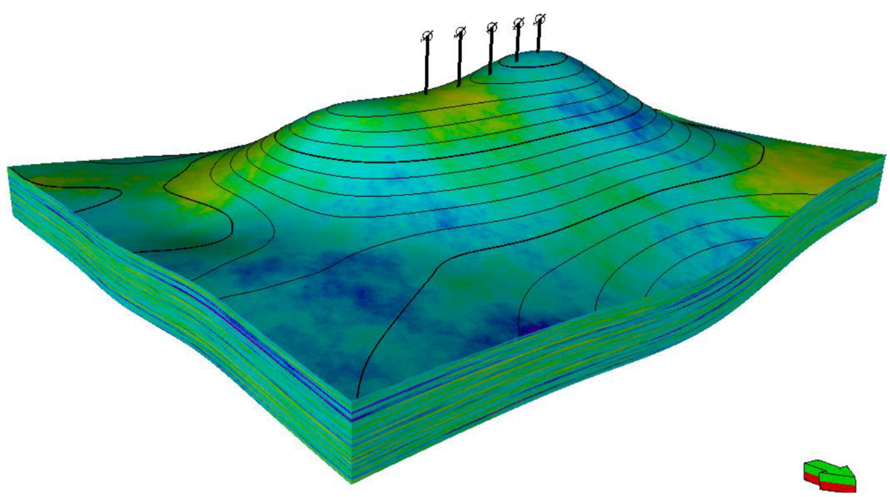

Figure 2—3D simulation model showing permeability distribution

A compositional fluid model, with modified Peng-Robinson equation of state (EOS), was used to calculate fluid properties. H_2 and other cushion gases (CO_2, CH_4, N_2) were considered as pure components and their EOS parameters (e.g., molecular weight, critical pressure, critical temperature, and acentric factor, etc.) were obtained from The Engineering ToolBox [13] and NIST Chemistry WebBook [14]. In addition, binary interaction coefficients were used to account for fluid properties of gaseous mixtures. The water phase was assumed to be a slightly compressible fluid with viscosity and density depending on the pressure change. The reservoir was assumed to be fully saturated with water initially and in an isothermal condition with a constant temperature value of 180°F. Hydrogen and cushion gases dissolution in the water phase was not considered in the study.

For rock physics flow parameters, the hydrogen-water relative permeability data, as shown in Figure 3, were adopted from experimental measurements presented by Alhotan et al., 2023 [15]. The reservoir rock compressibility was assumed to be 3.5 × 10^{-6} psi^{-1}. Table 1 summarizes the key input parameters used in this study.

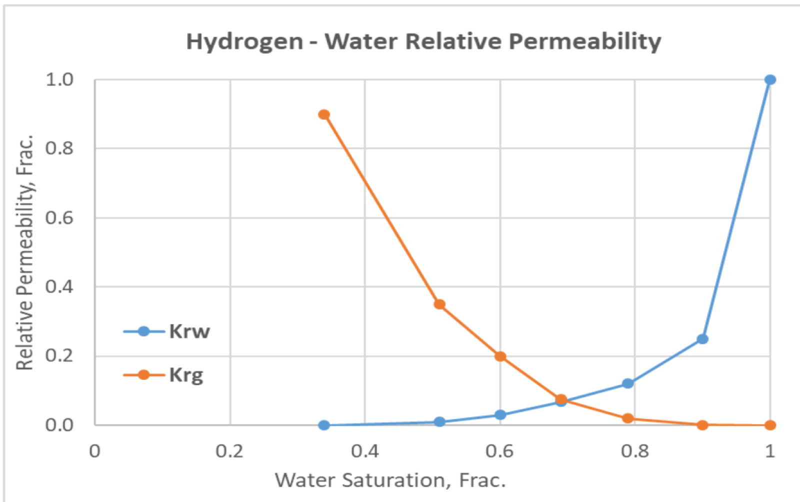

Figure 3—H_2-Water relative permeability curves used in the model

Table 1—The key synthetic reservoir model parameters

Parameter	Value
Avg. Porosity	0.2
Avg. Permeability	46 mD
kv/kh ratio	0.10
Initial Reservoir Pressure	2650 @ 5300 ft ssTVD
Water Density	63 lb/ft^3
Water Viscosity	0.49 cP
Water Compressibility	2.9 × 10^{-6} psi^{-1}
CO_2 Sp. Gravity	1.521
Irreducible water Saturation	0.34
Critical Gas Saturation	0.02

Reference Case Description

In the reference simulation case, five vertical wells were placed at the crest of the structure in order to utilize most of the available pore volume, as shown in Figure 4. All wells were completed in the top five layers of the reservoir. A generic operational scheme, as shown in Figure 5, of the underground hydrogen storage process was selected, which started with injection of cushion gas for two years followed by multiple cycles of hydrogen injection and production. In each cycle, gas rate and length of H_2 injection period are the same as in the H_2 production period, i.e., 10 mmscfd and 6 months, respectively. The injection of cushion gas and hydrogen was constrained by maximum bottom-hole pressure of 4000 psi. Correspondingly, during H_2 production periods, wells were constrained by the max. water production rate of 150 bbl/day and min. total gas (H_2+CG) production rate of 1.5 mmscfd to limit excessive production of formation water and cushion gas respectively. In addition, minimum bottom-hole pressure of 1500 psi was assigned to avoid issues with lifting the fluids to the surface. The production stage of the ninth and the final cycle was followed by an extended production period of hydrogen production for 4 years to recover the maximum quantity of hydrogen possible and deplete the aquifer pressure. The length of extended production period depends upon operational constraints like max. water production and min. purity level of produced H_2, etc.

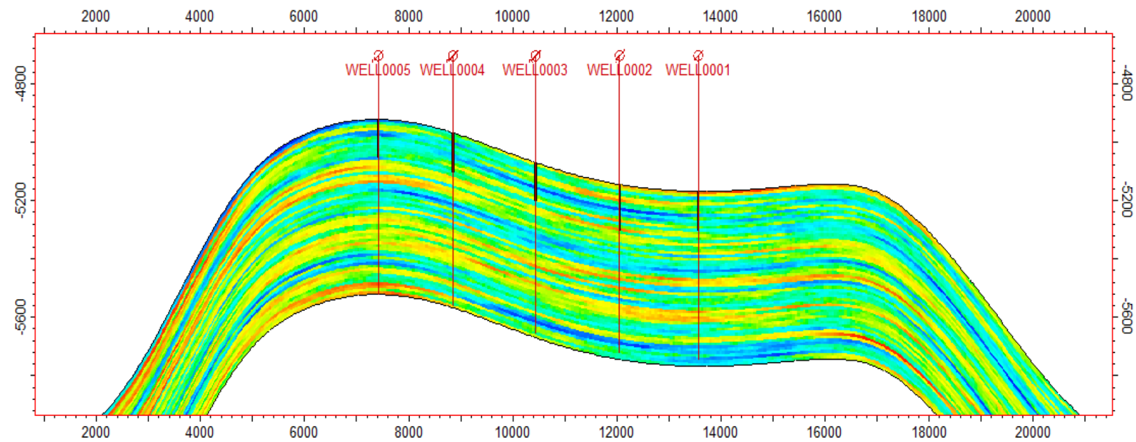

Figure 4—Model cross-section showing wells locations

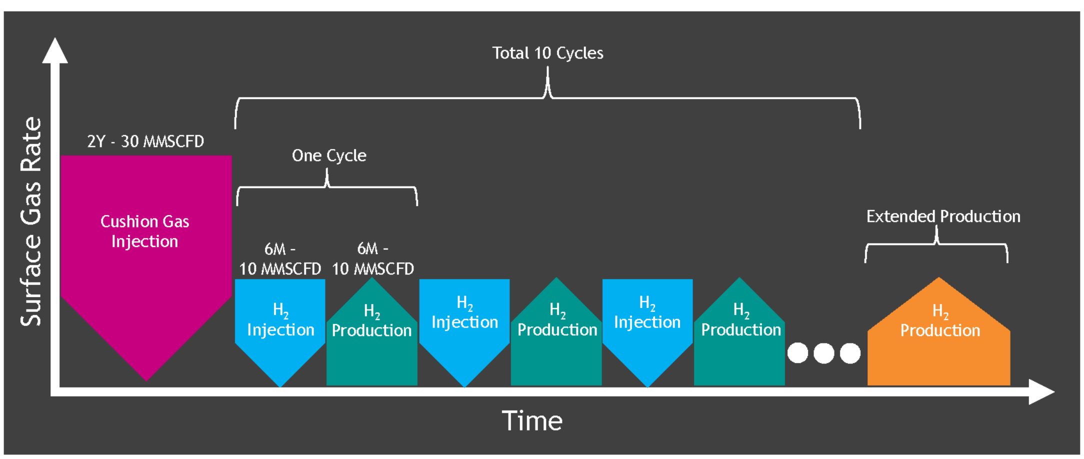

Figure 5—Illustration of operational scheme of H_2 injection in the reference case

Several simulation cases were performed in the following three categories considering the injection of three different types of cushion gas in different injected volumes, and different lengths of H_2 injection and production periods with multiple well configuration options.

1. Cushion Gas: Type of cushion gas (CH_4, N_2 and CO_2) and its injected volume requirement in the form of cushion gas to H_2 ratio (CGR)
2. Well Configuration: Well locations and placement
3. Operational Scheme: H_2 injection and production rates and periods

Simulation cases without any cushion gas injection were run in each category to establish a baseline for measuring the effectiveness of cushion gas injection. Then impact on key HUS performance parameters, such as storage capacity, reservoir pressure, produced H_2 purity and hydrogen recovery efficiency, was evaluated. All the simulation work was performed using Aramco's in-house compositional reservoir simulator called GigaPOWERS.

Impact of Cushion Gas

Cushion gas (CG) is needed to maintain reservoir pressure and hence to ensure H_2 deliverability. It also prevents trapping of hydrogen by water encroachment from the aquifer. CG was injected in different volumes for all three types and was presented in the form of a cushion gas ratio, which is defined as the total injected volume of CG prior to H_2 cycles divided by the total volume of H_2 planned to be injected in all cycles. In other words, a higher CGR means more injected volume of cushion gas. A CG injection period of 2 years was kept the same in all cases.

As shown in Figure 6, compared to no-CG injection, the reservoir pressure increases with CGR values for all three types of cushion gases. However, the magnitude of increase in reservoir pressure is higher in the case of lighter gases (N_2, CH_4) as compared to CO_2 as CG.

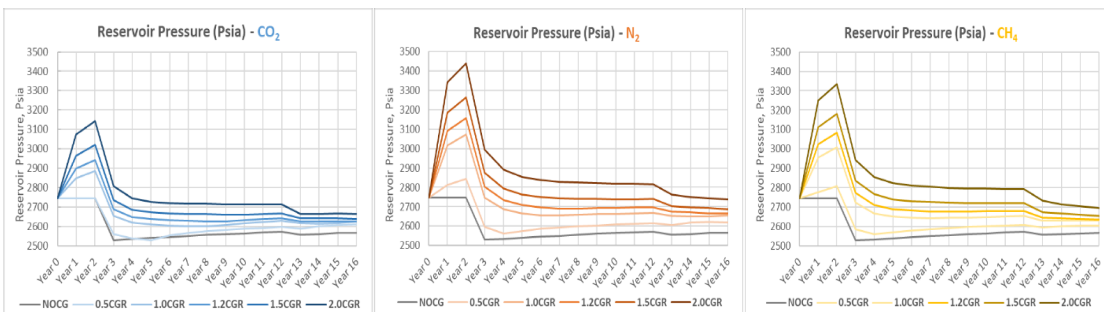

Figure 6—Change in reservoir pressure with different CGR values for all three types of cushion gases

Likewise, the total gas production in all CGR cases is higher than no-cushion gas injection case. For CGR values =>1.0, total gas production is almost the same during H_2 injection/production cycles due to max. gas production constraints; however, the effect of different CGR values is more pronounced during extended production periods when gas production depends upon energy left in the reservoir, as shown in Figure 7. This suggests that CG injected volume is not only required during H_2 cycles but is also an important factor during extended production periods to recover the remaining H_2.

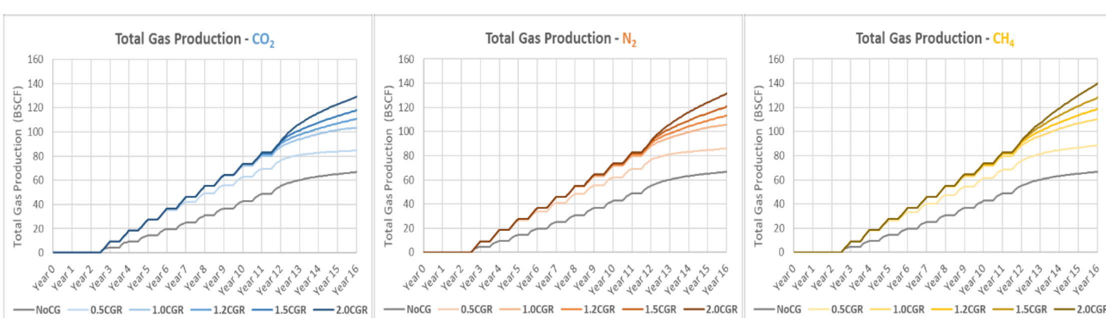

Figure 7—Change in total gas production with different CGR values for all three types of cushion gases

Hydrogen recovery efficiency (HRE) is defined as the ratio of total volume of H_2 injected divided by total volume of H_2 produced. As shown in Figure 8, regardless of the cushion gas type, H_2 recovery efficiency increases with increasing CGR; however, the optimum CGR ranges from 1.25 to 1.5. Beyond a CGR value of 1.5, there is no benefit observed from any CG type.

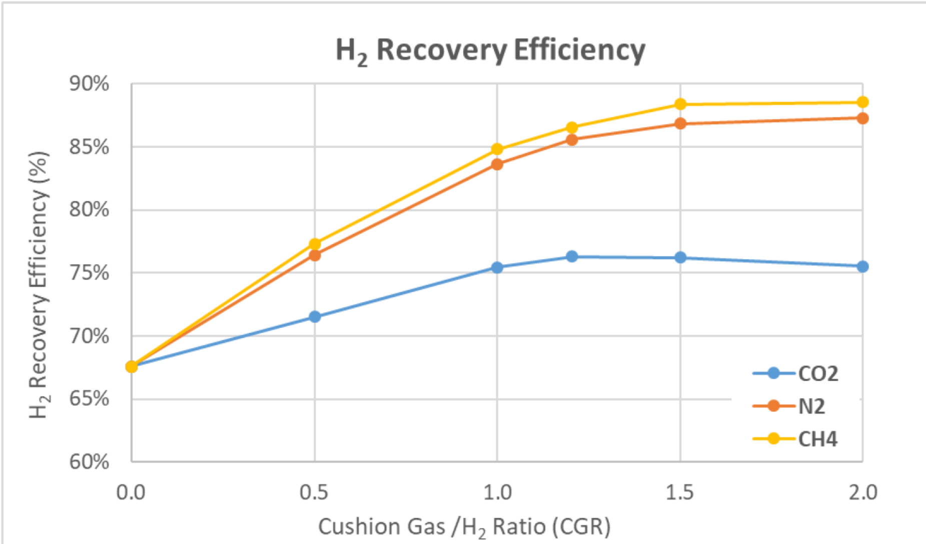

Figure 8—Change in hydrogen recovery efficiency (HRE) with CGR for all three types of cushion gases

To further investigate the effect of cushion gas type, no-cushion gas case and simulation cases of CGR=1.5 were compared for all three (CH_4, N_2 and CO_2) cushion gases. Figure 9 illustrates the impact of cushion gas type on the reservoir pressure.

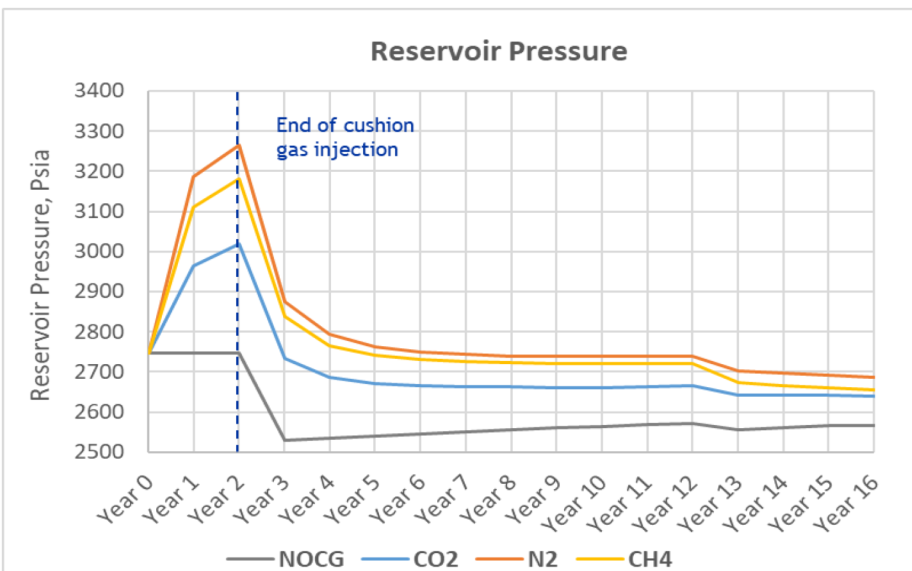

Figure 9—Change is reservoir pressure as a result of different types of cushion gas injection

Reservoir pressure builds up during the injection period of all types of cushion gas compared to no CG injection case. Injection of lighter cushion gases resulted in better aquifer pressurization, e.g., N_2 and CH_4 can increase the reservoir pressure to higher levels compared to CO_2 because of their higher Z-factors (Gas Compressibility Factor) under the reservoir thermodynamic conditions.

The reservoir storage capacity for hydrogen underground storage is defined as the total reservoir pore volume accessible by injection of cushion gas and is shown in Figure 10. Although the surface injection volume is the same for all CG cases, N_2 reservoir volume injection is higher due to its higher gas compressibility. This results in higher pressure buildup and also increased displacement of aquifer water, which ultimately results in higher storage capacity.

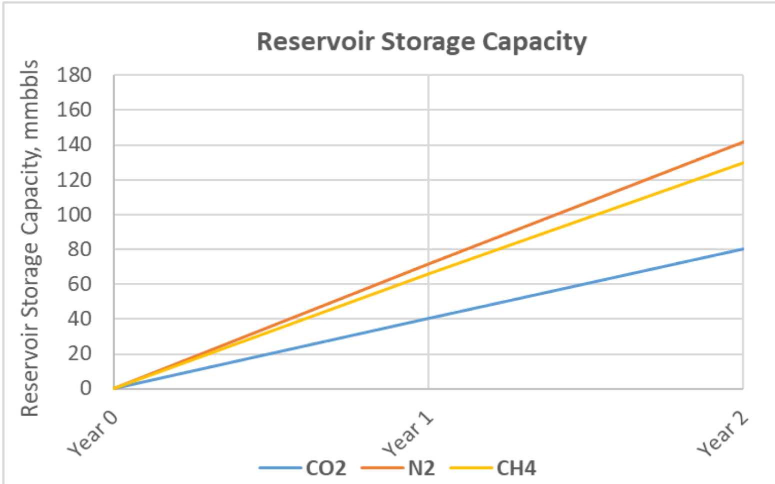

Figure 10—Reservoir storage capacity as a result of two years of injecting different cushion gases

Produced H_2 purity is defined as the ratio of H_2 production rate to total gas (H_2+CG) production rate. Produced hydrogen purity, in general, increases with increasing cycles in all the three cushion gas scenarios. This is explained by the increased hydrogen gas saturation occupying the near wellbore region hindering the cushion gas production. The production period in each cycle begins with very high H_2 purity (86-99%), which declines down to ~30-70% at the end of the cycle. During cycles, CH_4 and N_2 give better H_2 purity compared to CO_2. This is possible due to gas stratification as a result of density difference of CO_2 and H_2. At the end of H_2 cycles (during extended production period), H_2 purity gradually declines due to less H_2 being available around the wellbore. CG production increases during this phase and lighter CG, CH_4, which in this case is produced more and hence results in the least purity for produced hydrogen. Figures 11 and 12 illustrate the effect of different types of cushion gases on the purity of produced H_2.

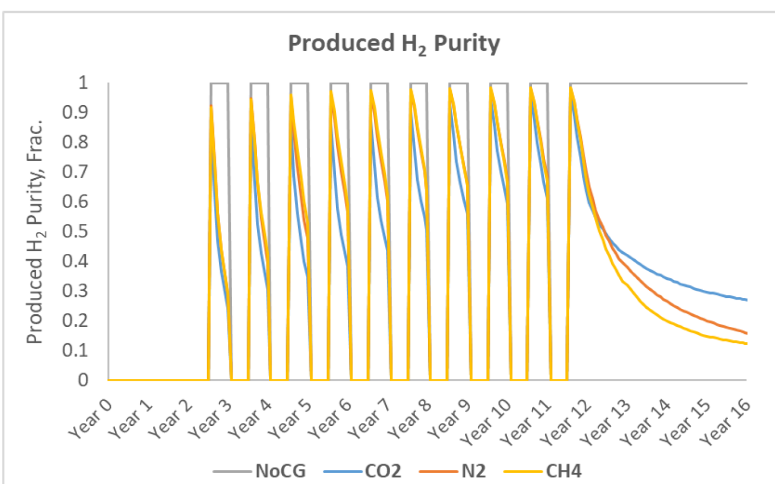

Figure 11—Produced H_2 purity for different types of cushion gases

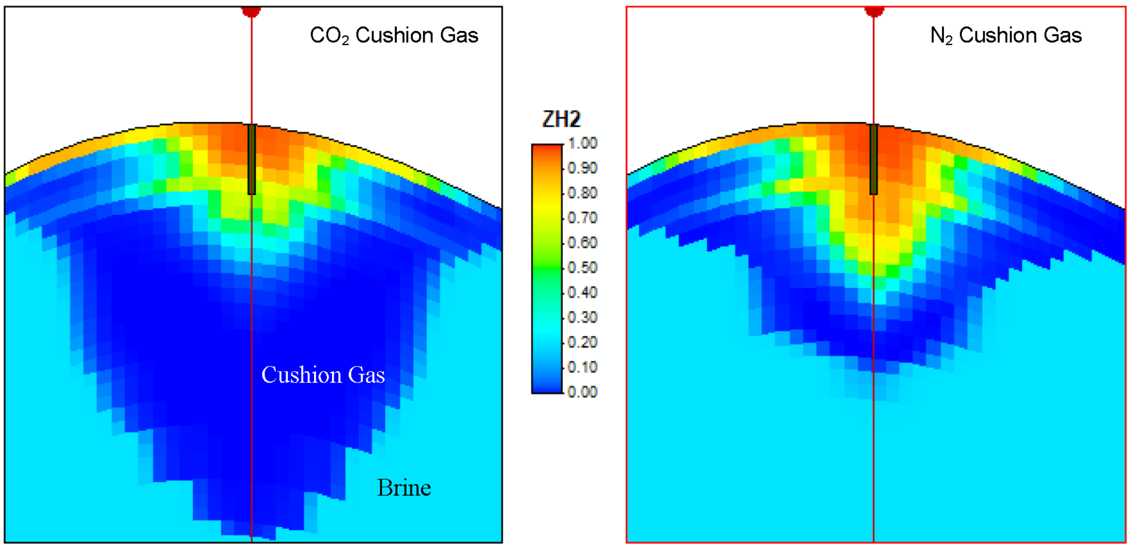

Figure 12—Effect of gas segregation due to density difference between H_2 and cushion gases (N_2 and CO_2)

Hydrogen recovery efficiency (HRE), as shown in Figure 13, increases with increasing the number of cycles in all CG cases due to more H_2 accumulation around the wells. No-cushion gas gives the lowest recovery due to lack of the pressure support necessary for H_2 deliverability. Lighter CG injection such as CH_4 and N_2 resulted in better recovery efficiency compared to CO_2 due to better pressure support and reduced gas segregation effect.

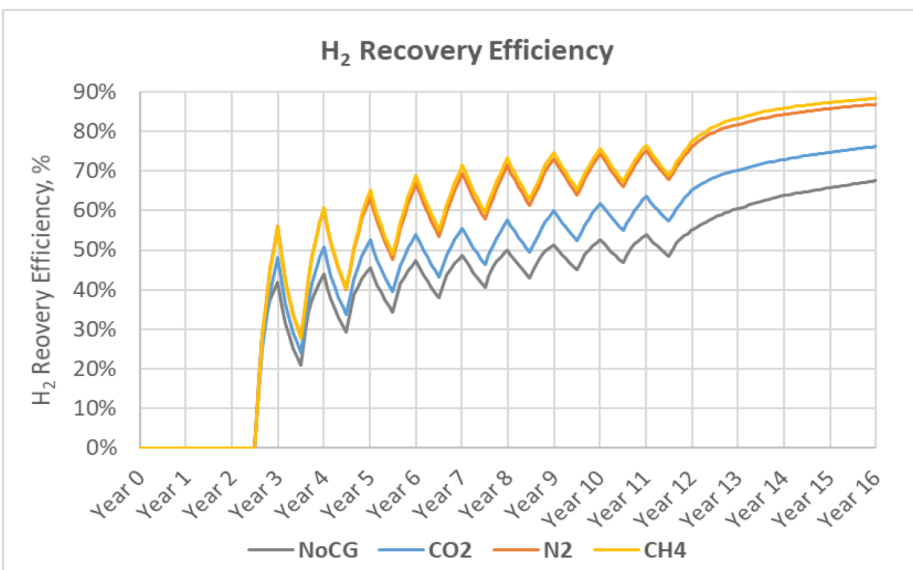

Figure 13—Hydrogen recovery efficiency with H_2 cycles for all types of cushion gases

Impact of Well Configuration

Well configuration is comprised of two elements, i.e., well completion and well location. For well completion sensitivity, three scenarios were considered where wells were completed in the top 5, 10 and 15 layers of the reservoir. These three completion scenarios were evaluated for no-cushion gas and for all three types of cushion gases. As shown in Figure 14, regardless of CG type, wells completed in the top five layers resulted in the highest H_2 recovery efficiency.

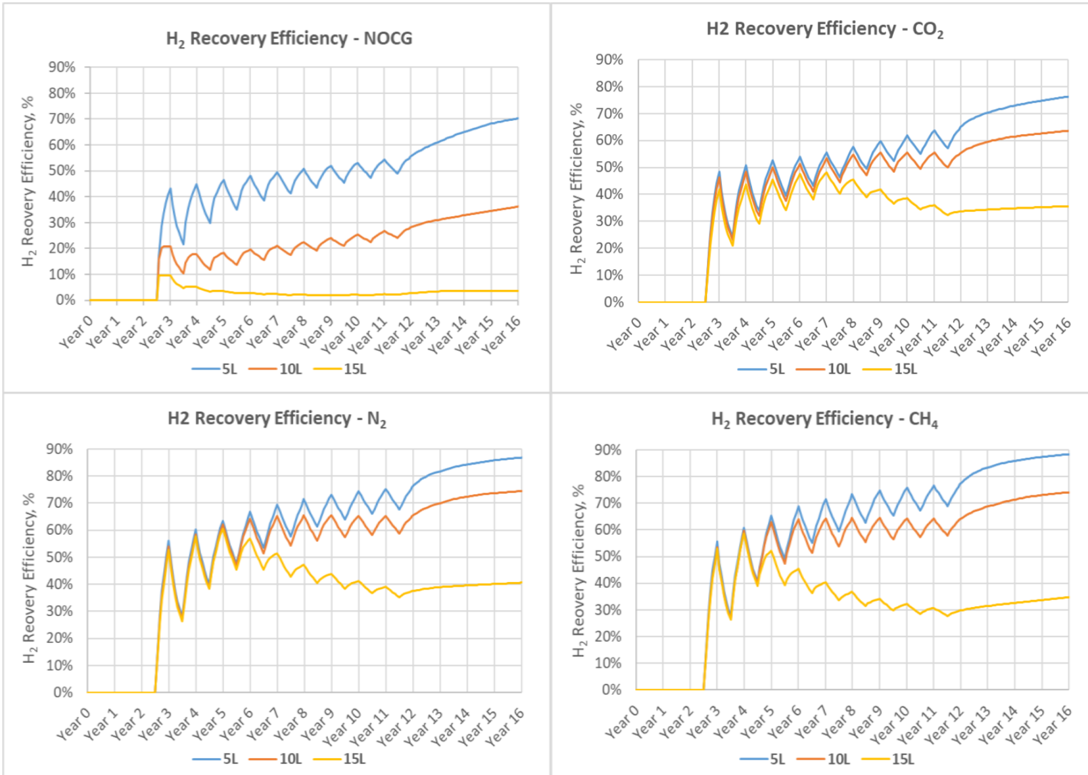

Figure 14—Effect of three different well completions on H_2 recovery efficiency for all types of cushion gases

This is explained by less water and CG production due to the effect of gas gravity segregation as illustrated in Figure 15. For the same completion type, for example the top five-layer (5L) scenario, the lighter cushion gases (CH_4, N_2) tend to perform better compared to no-CG and CO_2 cases.

For well location sensitivity, performance of an updip well (WELL0005) was compared with a downdip well (WELL0001). The gas-water gravity segregation effect was found to be dominated as gas tends to flow toward the top of the structure that caused early water coning in the downdip well, as shown in Figure 16. This was also confirmed by comparison of the gas-water ratio (WGR) of updip and downdip wells, which resulted in higher WGR values for the downdip well, as shown in Figure 17. As a result, H_2 recovery efficiency of the updip crestal well was found to be better than the downdip well.

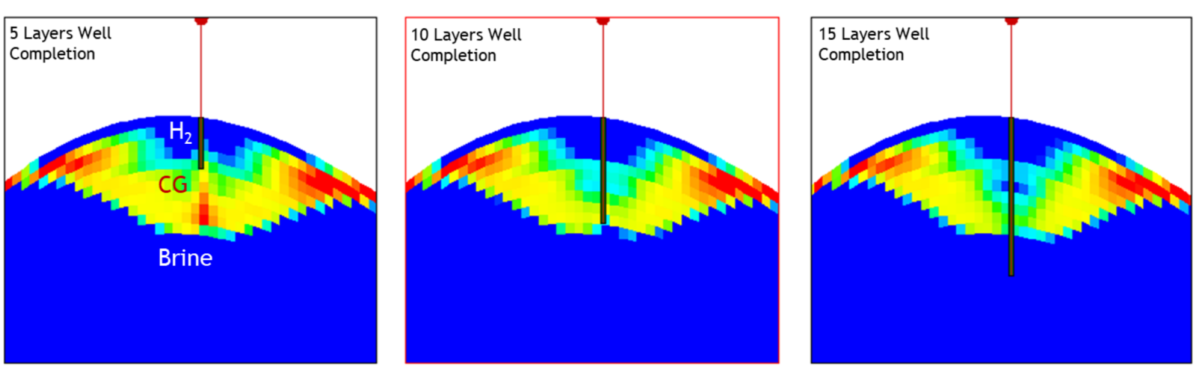

Figure 15—Illustration of water and CG encroachment in three well completion types for CO_2 cushion gas injection

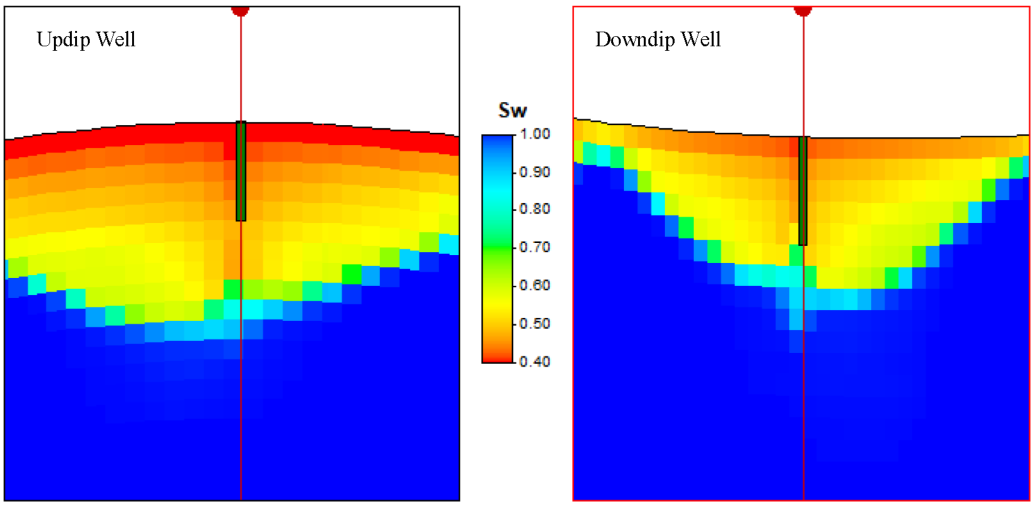

Figure 16—Comparison of water encroachment in updip and downdip wells for CO_2 injection cases

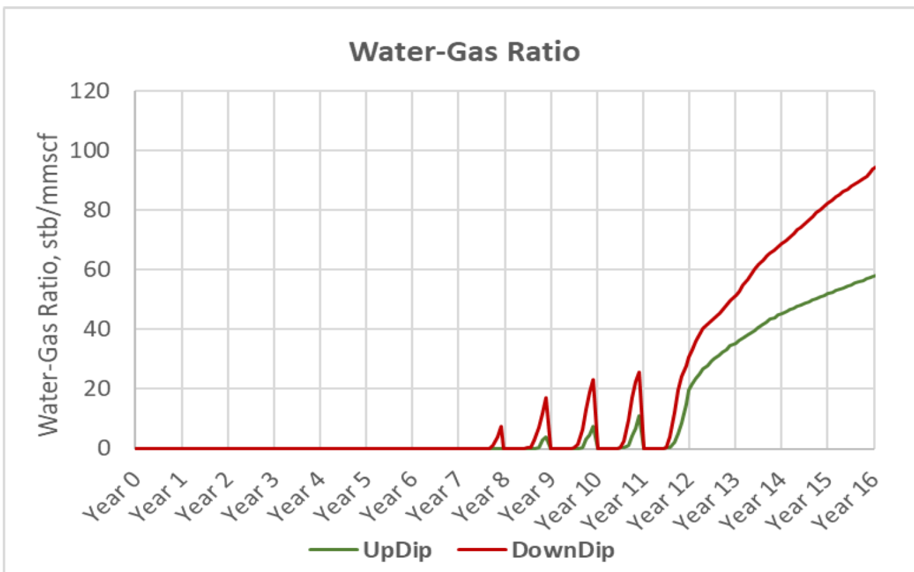

Figure 17—Water-gas ratio comparison of updip and downdip wells

Impact of Operational Scheme

Operational scheme parameters include the length of the H_2 gas injection and production period as well as the H_2 injection and production rates. Three different operational scheme designs, as shown in Figure 18, were evaluated. The length of injection and production period was different with different rates. However, the length of each cycle (1 year) and total injection and production volumes of H_2 in each cycle were kept the same in all three scenarios. The same volume of CG was injected prior to H_2 cycles in all cases.

In design-A, the length of the H_2 production period in each cycle was the same as the H_2 injection periods, i.e., 6 months. The H_2 production rate was equal to the H_2 injection rate, i.e., 10 mmscfd.

In design-B, the length of the H_2 production period in each cycle was shorter than H_2 injection periods, i.e., 4 months and 8 months, respectively. The H_2 production rate was higher than the H_2 injection rate, i.e., 15 and 7.5 mmscfd, respectively.

In design-C, the length of the H_2 production period in each cycle was greater than the H_2 injection periods, i.e., 8 months and 4 months, respectively. The H_2 production rate was less than the H_2 injection rate, i.e., 7.5 and 15 mmscfd, respectively. All three operational scheme designs were evaluated for no-cushion gas and for all three types of cushion gases.

Regardless of CG type, H_2 recovery efficiency, as shown in Figure 19, was better in case of design-C where the production period was longer with a lower daily withdrawal rate. The lower H_2 withdrawal rate and, therefore, the lower drawdown resulted in less CG and water encroachment in design-C. Nevertheless, recovery efficiency increases with cycles in all designs. Figure 20 shows the results of operational scheme design sensitivity for produced H_2 purity when CO_2 was used as a cushion gas. Produced H_2 purity was higher in design-C; however, purity increases with the number of cycles in all three design cases. The same behavior was observed when N_2 and CH_4 were used as cushion gas.

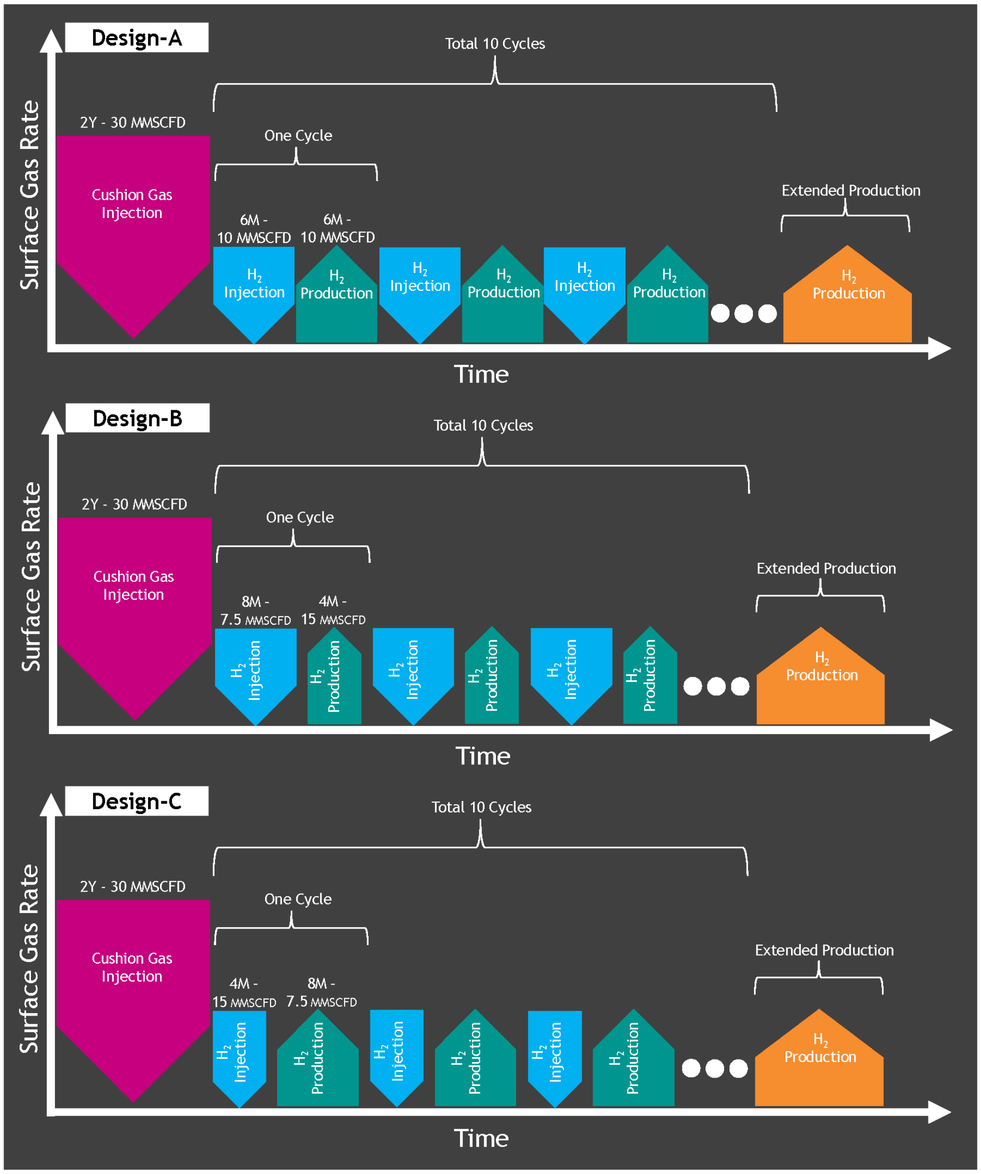

Figure 18—Three designs of the H_2 operational scheme were considered in this study

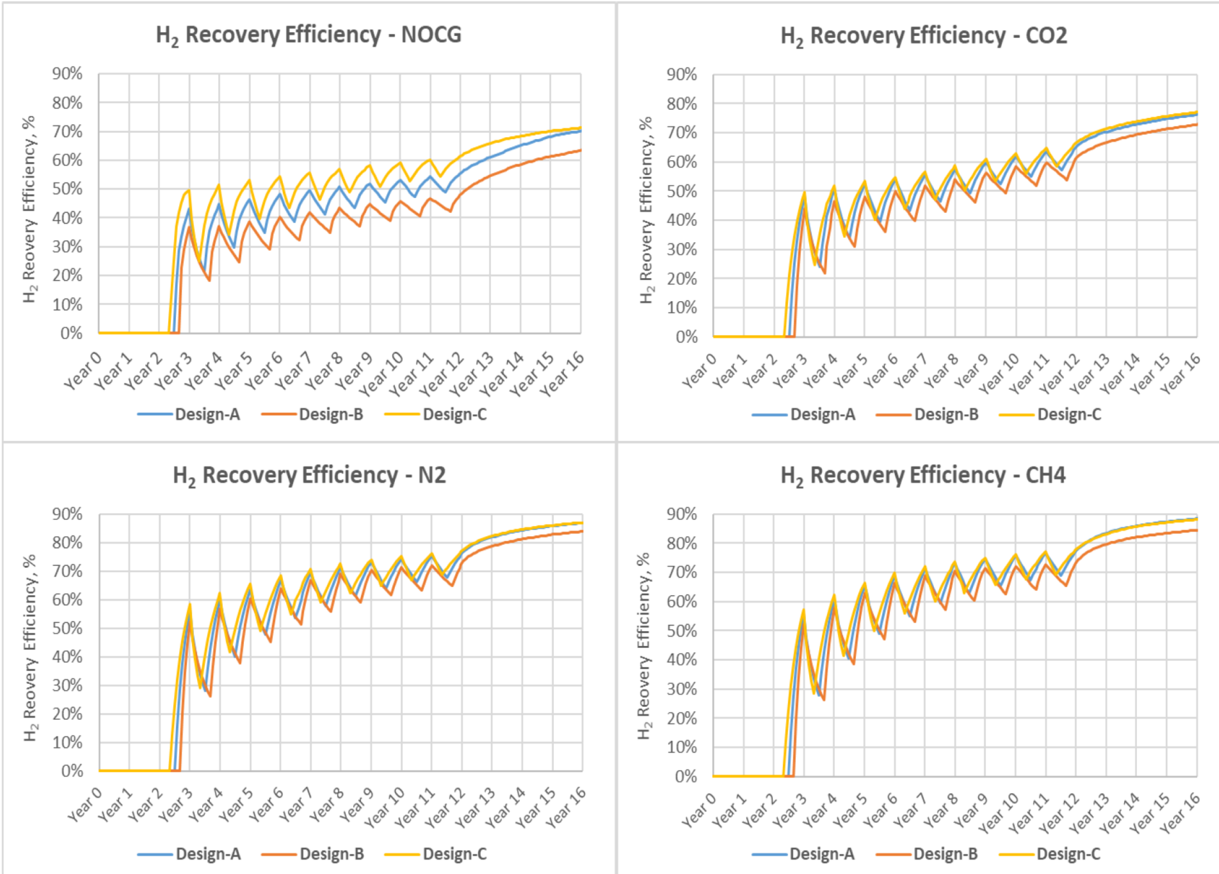

Figure 19—Impact of different designs of operational scheme on H_2 recovery efficiency

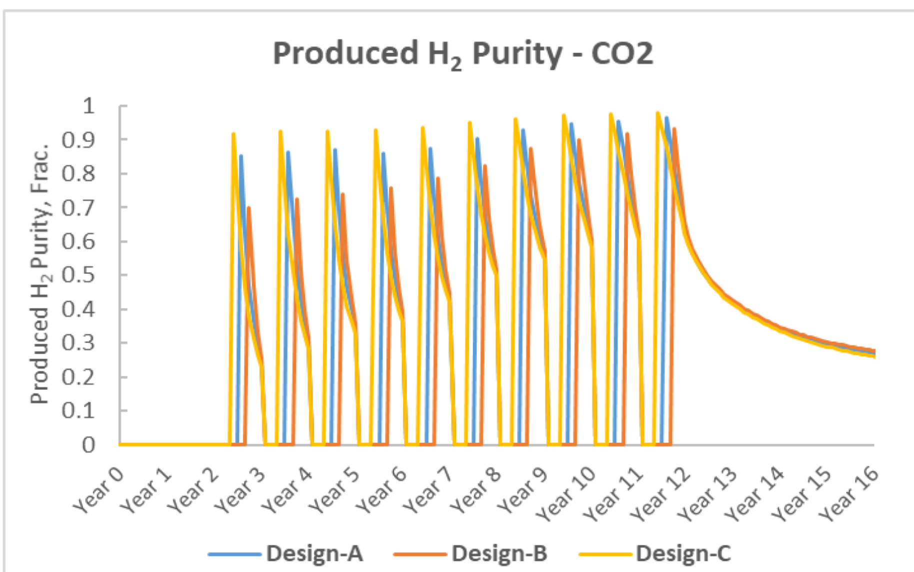

Figure 20—Impact of different designs of operational scheme on produced H_2 purity

Conclusions

The following key points can be derived from the simulation results of this study:

- Injection of cushion gas prior to H_2 injection/production cycles is needed to build reservoir pressure, which ensures H_2 deliverability and improves H_2 recovery efficiency.
- Regardless of the cushion gas type, H_2 recovery efficiency increases with increasing the cushion gas-hydrogen ratio (CGR); however, the optimum CGR ranges from 1 to 1.5. Beyond the CGR value of 1.5, there is no additional benefit.
- Individual cycle recovery efficiency increases with time for all types of cushion gases and also in the case of no-cushion gas. This is possibly due to increased hydrogen gas saturation occupying the near wellbore region.
- N_2 injection due to its higher compressibility resulted in the highest reservoir pressure buildup followed by CH_4 and CO_2.
- Lighter cushion gas (N_2 and CH_4) injection provided better recovery efficiency than CO_2 due to less lateral gas movement and viscous fingering.
- Produced hydrogen purity increases by increasing the number of cycles in all the three cushion gas scenarios. CH_4 and N_2 gives better H_2 purity compared to CO_2. During an extended production period, H_2 purity gradually declines due to less H_2 being available around the wellbore.
- Hydrogen wells placed updip and completed in the top part of the reservoir resulted in better performance in terms of H_2 recovery and purity in all types of cushion gas scenarios due to reduced cushion gas production and less water encroachment.
- HUS performance depends upon operational scheme design parameters, such as H_2 injection and withdrawal rates and length of injection/production period. The operational design with lower and longer withdrawal rates tends to result in better H_2 recovery efficiency and produced H_2 purity.
- The results of this study were based on a simplistic reservoir description, hence real reservoir cases may present unique/different challenges.

In conclusion, this work contributes to providing better understanding of key design parameters that can affect performance of the HUS process. It can facilitate designing optimal operational strategies to enhance the efficiency and reliability of hydrogen underground storage projects.

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